

Lab 11.1: Deep Dive in Pack Analysis – Using Wireshark and Network Miner

Remove Lab Requirement Instructions in your submission

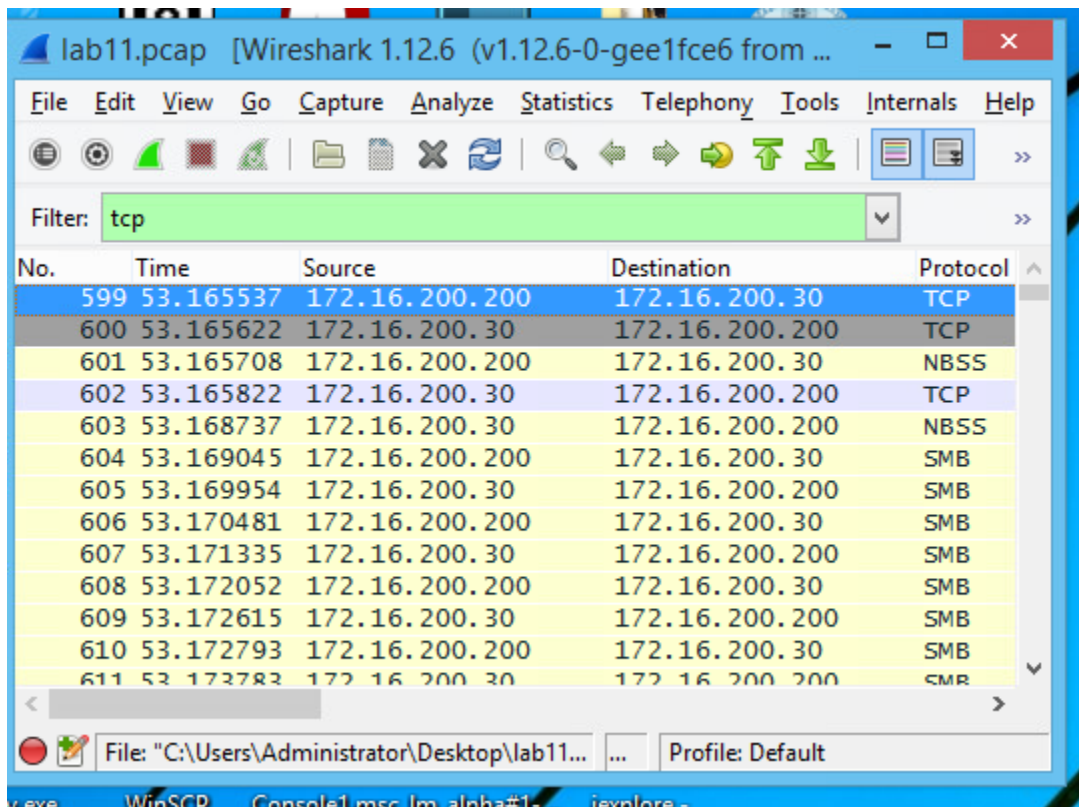
Lab Requirements:

1. In this assignment you will complete the assigned labs according to the instructions. This document will serve as a template to fill in answers where noted and screenshots required for each question.
2. Please make sure your screenshots follow the requirements for the course. You do not want to lose points needlessly for improper screenshots or formatting.
3. **Please make sure you show the syntax in your screenshots!** In order to get credit for your screenshots, you must show syntax and the output.
4. It is expected that any and all external sources used to complete any labs or support answers are properly cited in APA format and in a References Section at the end of the lab document.
5. Please name the file Lastname_LabTitle_Week# using your own last name at the beginning of the filename, lab title, and the appropriate week number. For example, if I were to name this lab: **Allen_Lab11.1_Week11.pdf**
6. Submit your work to the appropriate Assignment/Activity link in Blackboard. Do not submit assignments via course messages.
7. Points are assigned based on the completeness of your answers. Points will be deducted for errors and omissions.

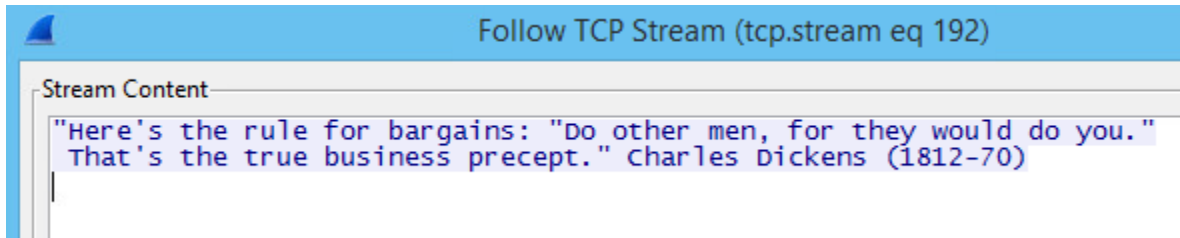
8. There will be challenge flags throughout these labs. These are not required to be completed for the assignments, but feel free to complete them on your own. Also feel free to go through the labs as many times as you want. Going through them a couple of times can help students understand the material.

Lab 11.1: Part 1 – Viewing Protocols with Wireshark

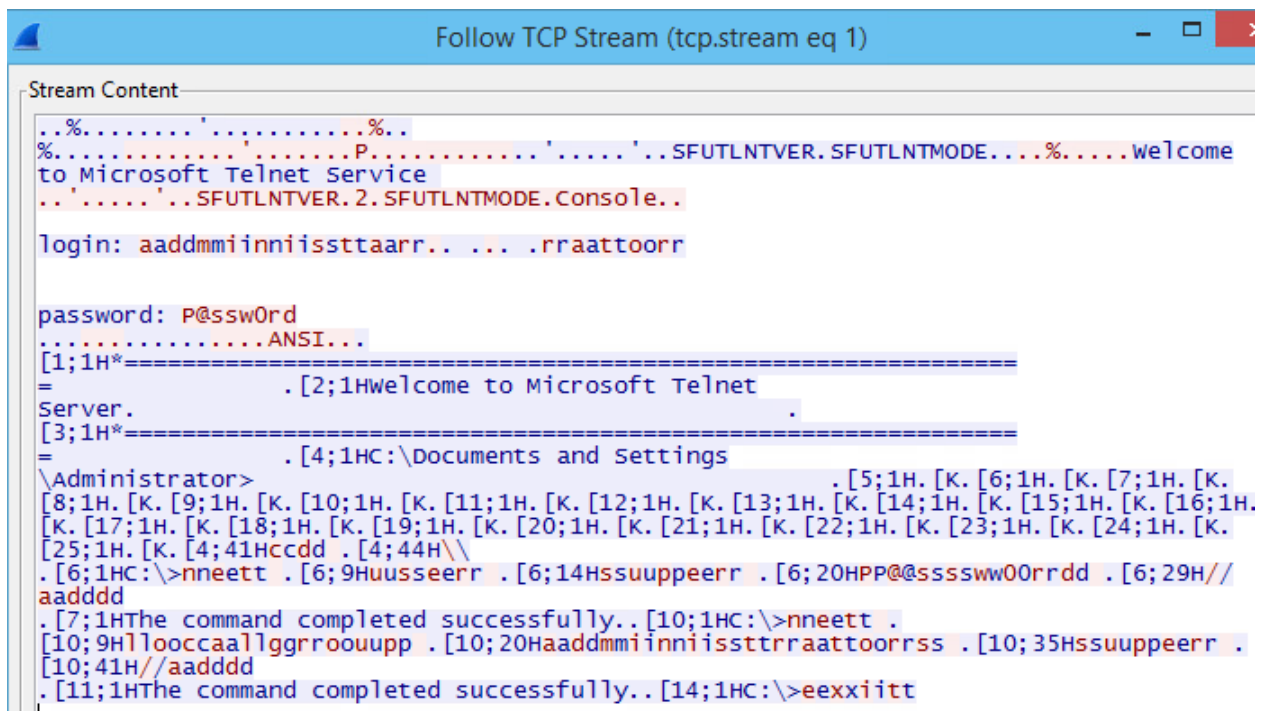
1. Question 9



2. Question 16



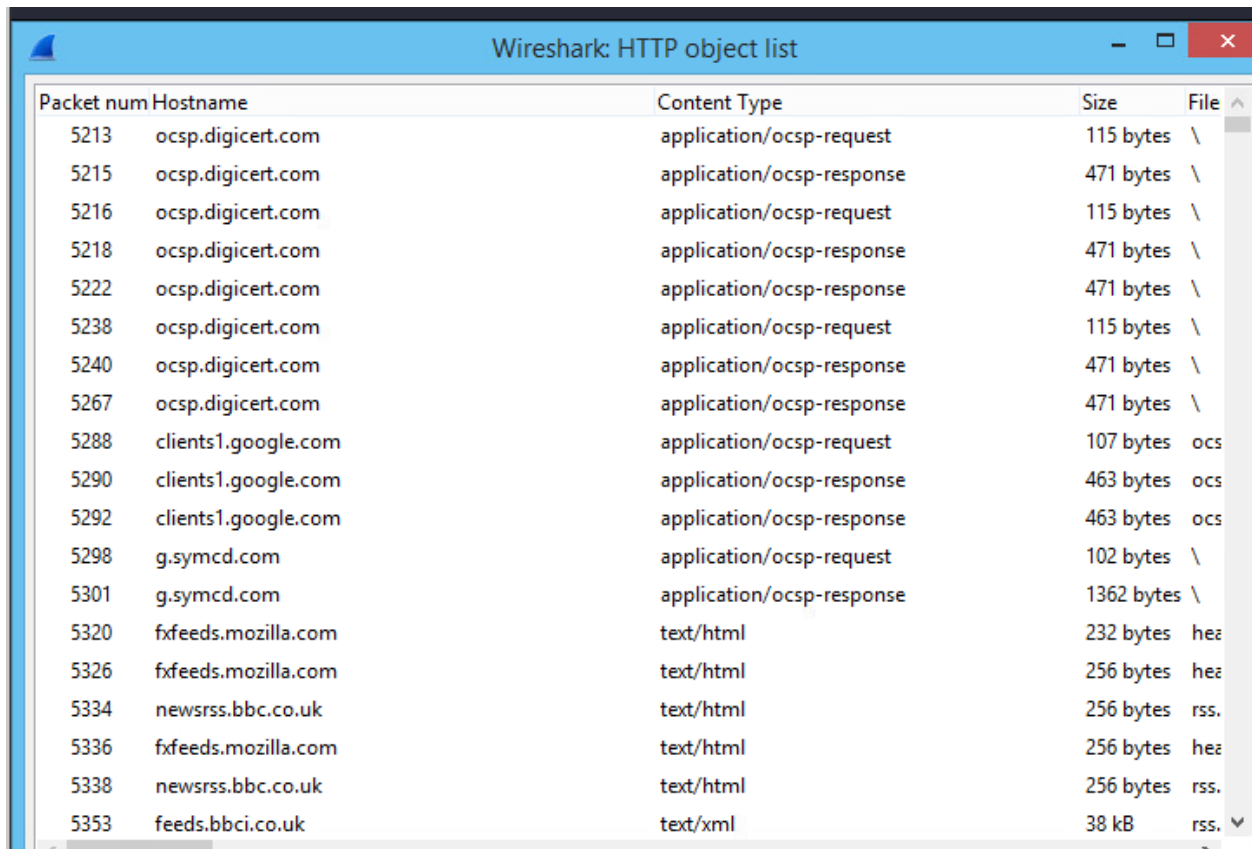
3. Question 18



4. Question 20

Lab 11.1: Part 2 – Parsing Objects with Wireshark

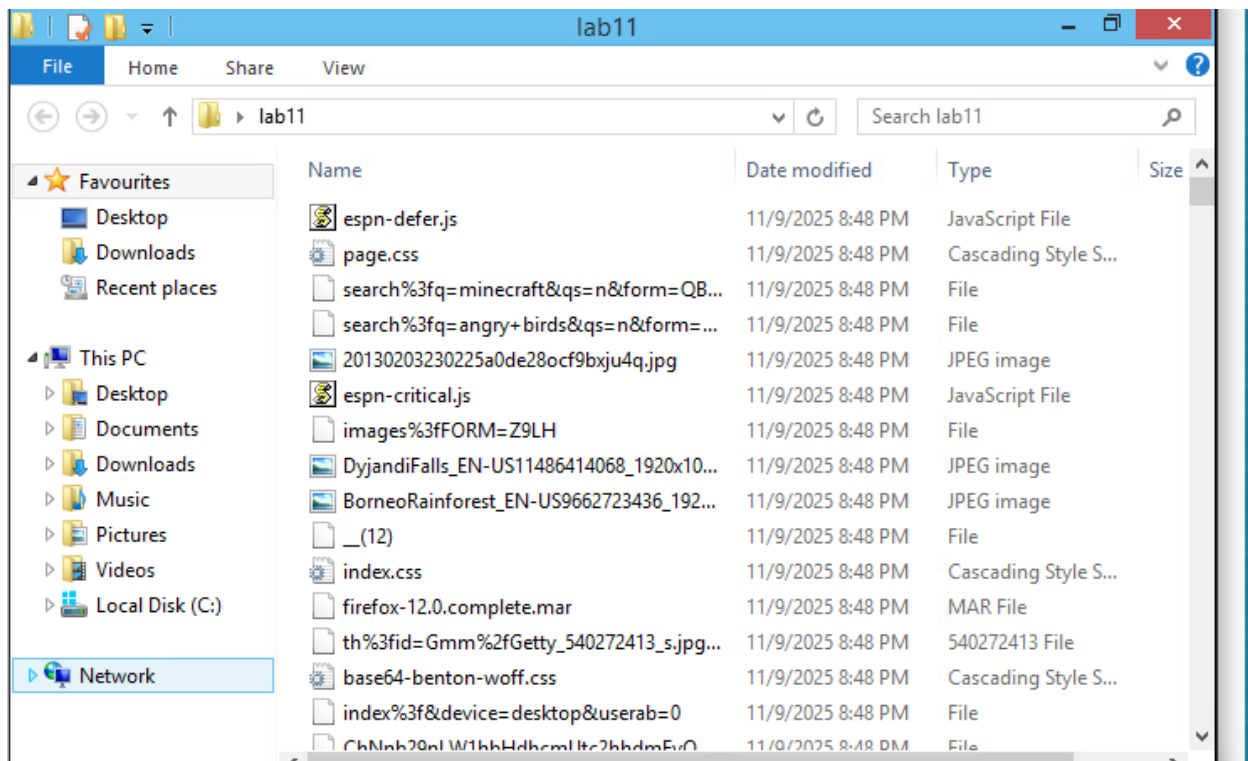
1. Question 2



The image shows a screenshot of the 'Wireshark: HTTP object list' window. It contains a table with five columns: 'Packet num', 'Hostname', 'Content Type', 'Size', and 'File'. The table lists various network objects, including OCSP requests and responses from digicert.com, clients1.google.com, and g.symcd.com, as well as HTML and XML feeds from mozilla.com, bbc.co.uk, and bbc.co.uk.

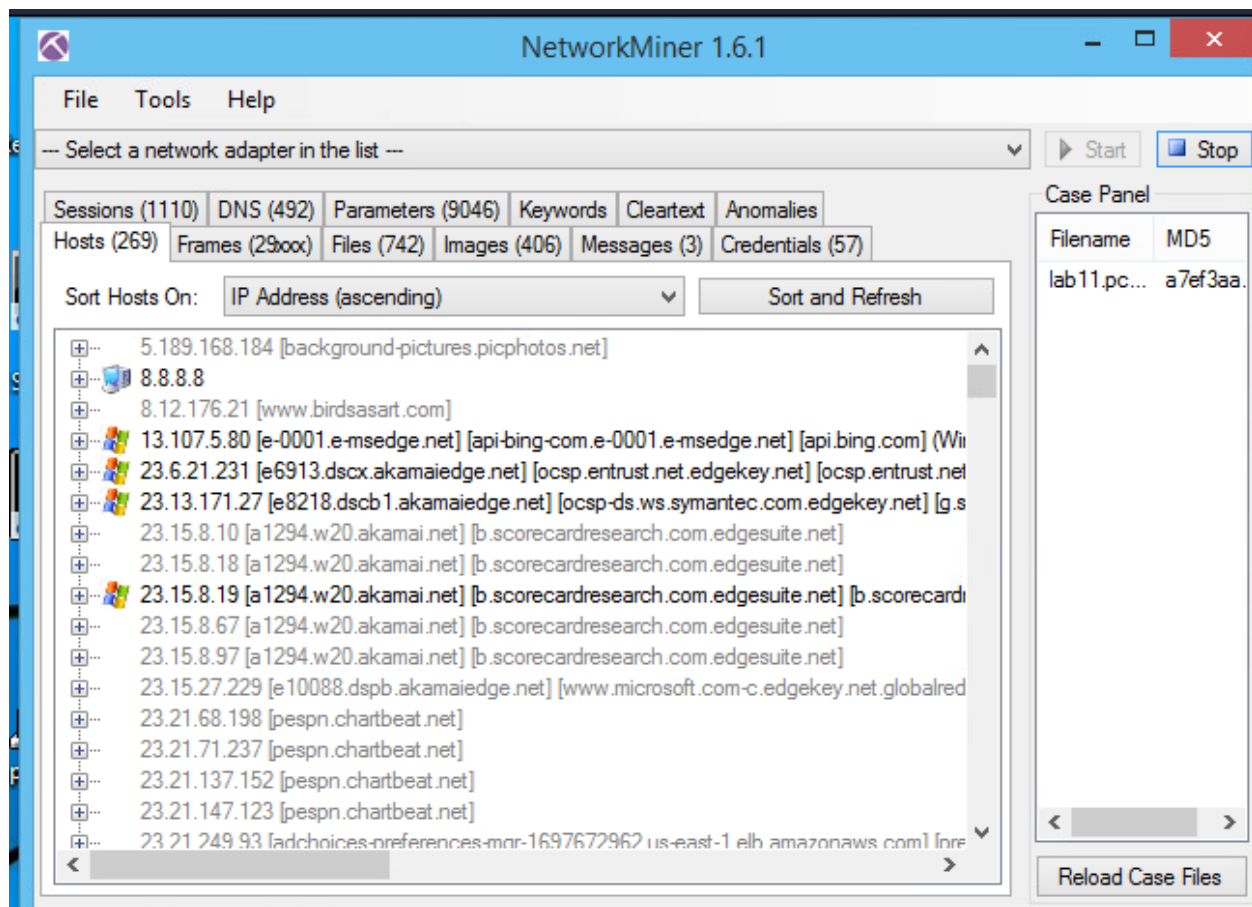
Packet num	Hostname	Content Type	Size	File
5213	ocsp.digicert.com	application/ocsp-request	115 bytes	\
5215	ocsp.digicert.com	application/ocsp-response	471 bytes	\
5216	ocsp.digicert.com	application/ocsp-request	115 bytes	\
5218	ocsp.digicert.com	application/ocsp-response	471 bytes	\
5222	ocsp.digicert.com	application/ocsp-response	471 bytes	\
5238	ocsp.digicert.com	application/ocsp-request	115 bytes	\
5240	ocsp.digicert.com	application/ocsp-response	471 bytes	\
5267	ocsp.digicert.com	application/ocsp-response	471 bytes	\
5288	clients1.google.com	application/ocsp-request	107 bytes	ocs
5290	clients1.google.com	application/ocsp-response	463 bytes	ocs
5292	clients1.google.com	application/ocsp-response	463 bytes	ocs
5298	g.symcd.com	application/ocsp-request	102 bytes	\
5301	g.symcd.com	application/ocsp-response	1362 bytes	\
5320	fxfeeds.mozilla.com	text/html	232 bytes	hea
5326	fxfeeds.mozilla.com	text/html	256 bytes	hea
5334	newsrss.bbc.co.uk	text/html	256 bytes	rss.
5336	fxfeeds.mozilla.com	text/html	256 bytes	hea
5338	newsrss.bbc.co.uk	text/html	256 bytes	rss.
5353	feeds.bbc.co.uk	text/xml	38 kB	rss. v

2. Question 8

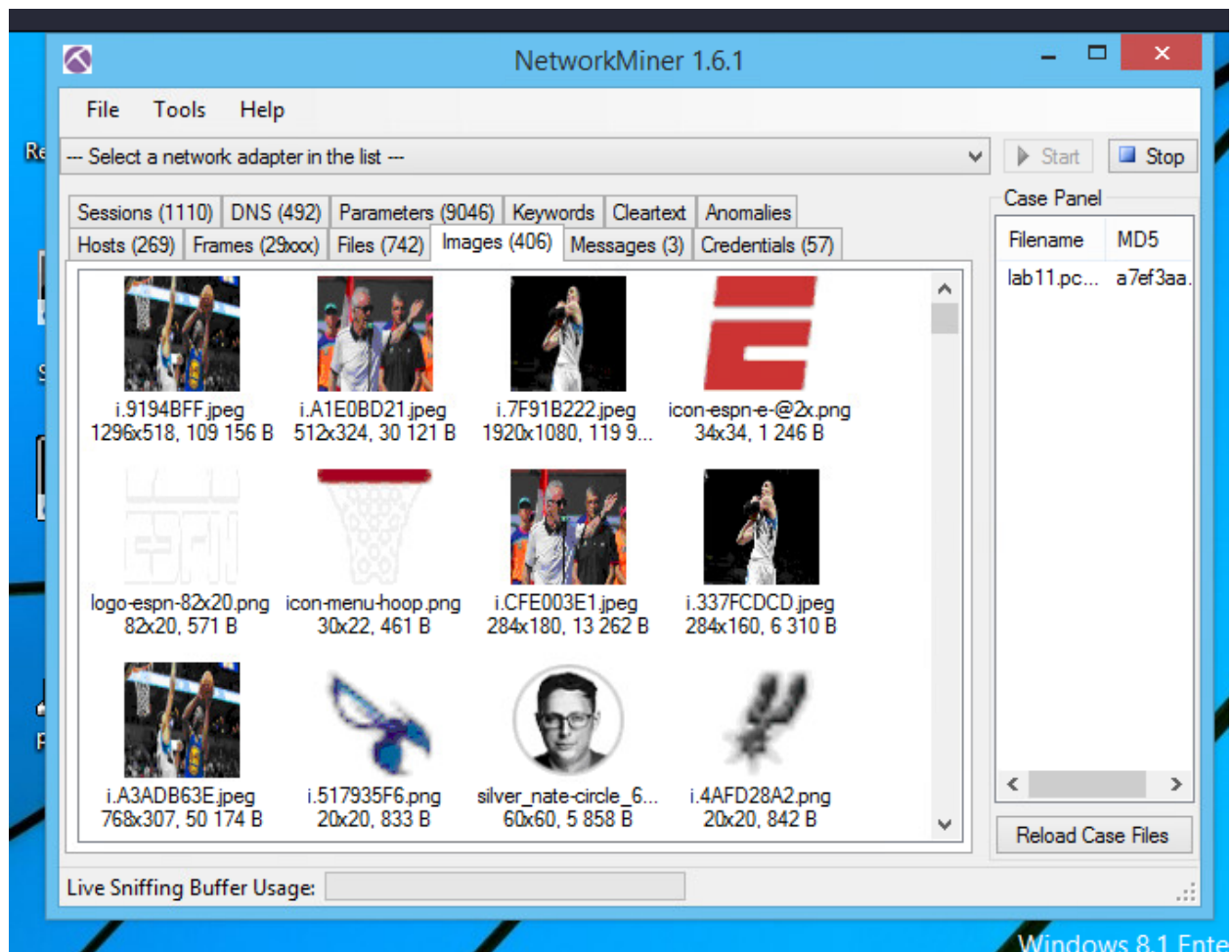


Lab 11.1: Part 3 – Using Network Miner

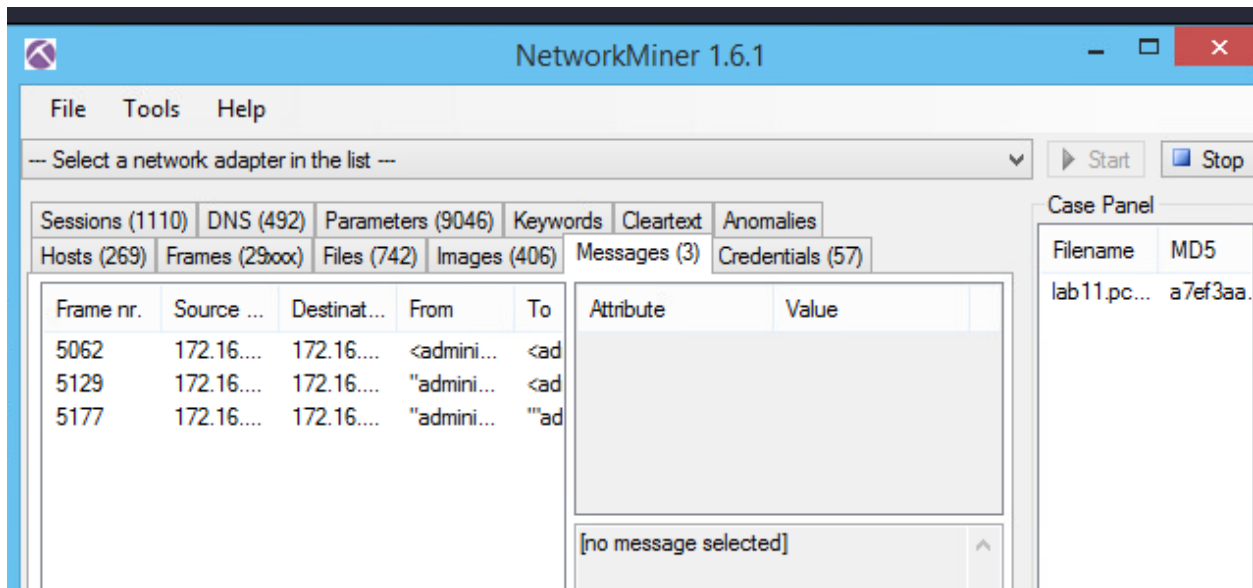
1. Research and describe what **NetworkMiner** does
2. Question 3



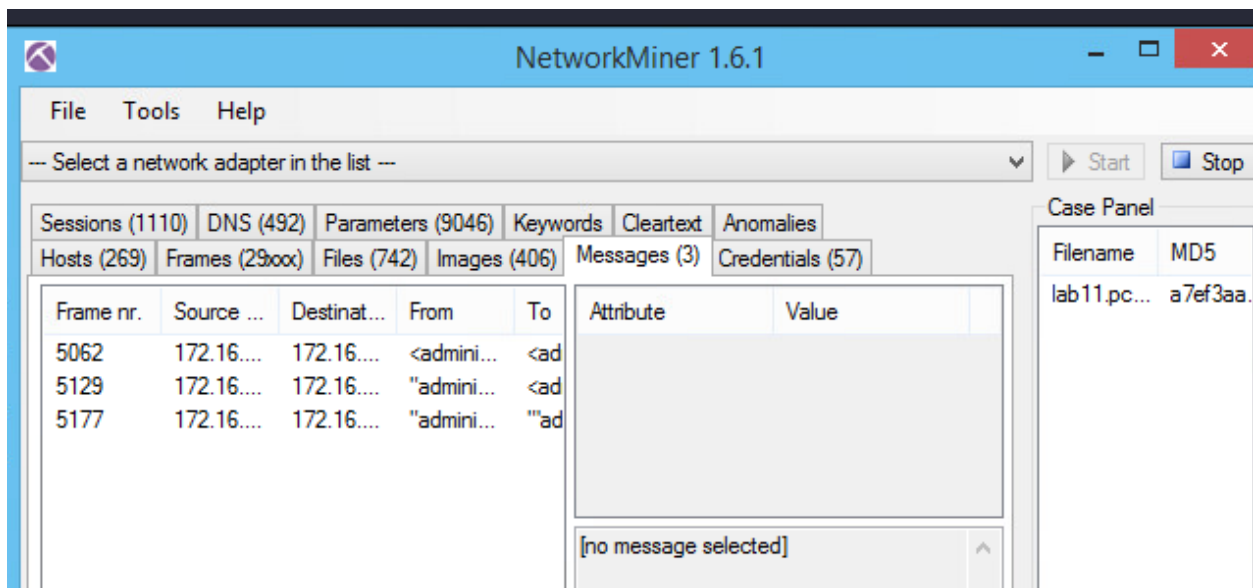
3. Question 4



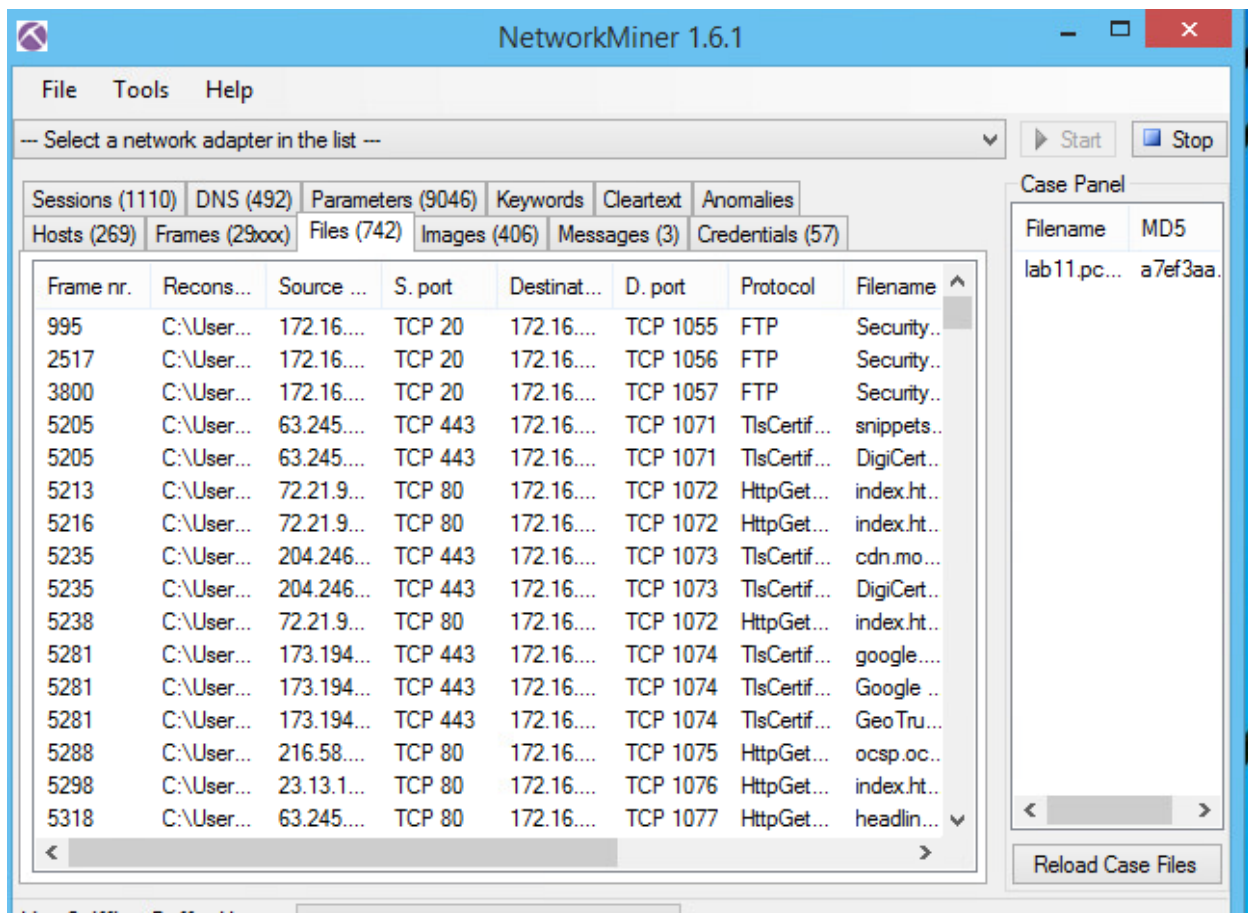
4. Question 5



5. Question 6



6. Question 7



Lab 11.1: Part 4 – Lessons Learned

This lab reinforced the critical role of packet analysis in network security and incident response. Using Wireshark and NetworkMiner deepened my understanding of how protocols behave in real-time and how forensic artifacts can be extracted from captured traffic. I gained practical insight into identifying suspicious patterns, reconstructing sessions, and parsing objects such as images and executables from PCAP files. The hands-on experience clarified how metadata and payloads reveal user activity, which is essential for threat hunting and post-incident analysis. I was surprised by the granularity of information available in packet captures—especially how much can be inferred from seemingly benign traffic. One challenge was filtering relevant packets efficiently in Wireshark; crafting precise display filters took trial and error. NetworkMiner’s ability to extract files and credentials was impressive, but it also highlighted how vulnerable unencrypted traffic can be. Ensuring proper screenshot syntax and formatting for submission also required attention to detail.